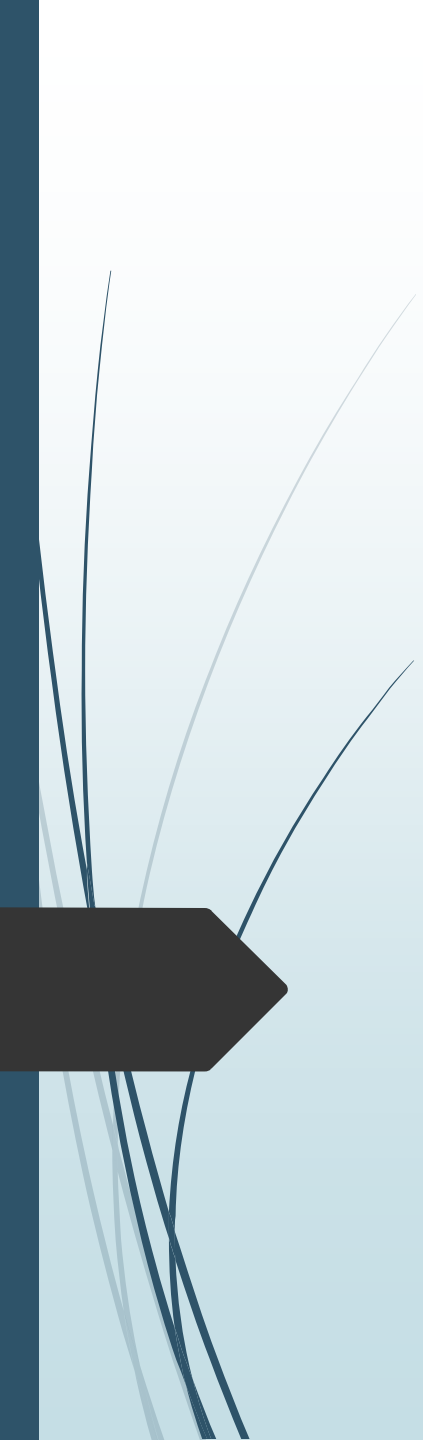




Small-Language Multi-Agent Debate to Identify Safety Attacks

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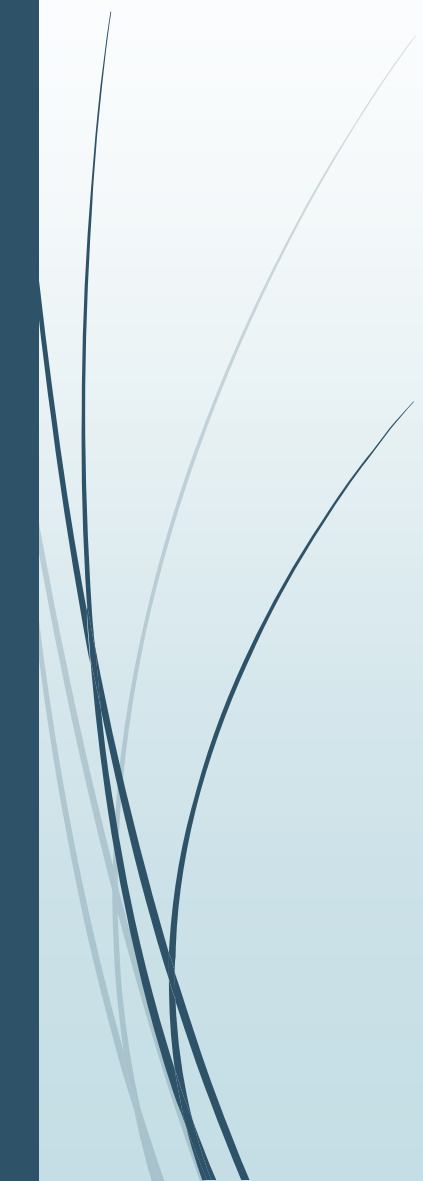
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Abstract

- This proposal introduces Small-Language Multi-Agent Debate (SLMAD), a conceptual framework designed to detect safety issues in questions and question-answer (Q-A) pairs.
- SLMAD leverages debate among small language models (SLMs) to collaboratively assess whether an input or output is safe.
- The system aims to provide interpretable, cost-effective safety evaluation for large language models (LLMs).



Background and Context

- LLMs have achieved significant breakthroughs in NLP tasks but remain vulnerable to safety attacks and misuse.
- Attacks such as data poisoning, jailbreak prompts, and harmful query generation challenge the safety alignment of these models.
 - References: Ouyang et al. (2022), Bai et al. (2022), Pang et al. (2024)

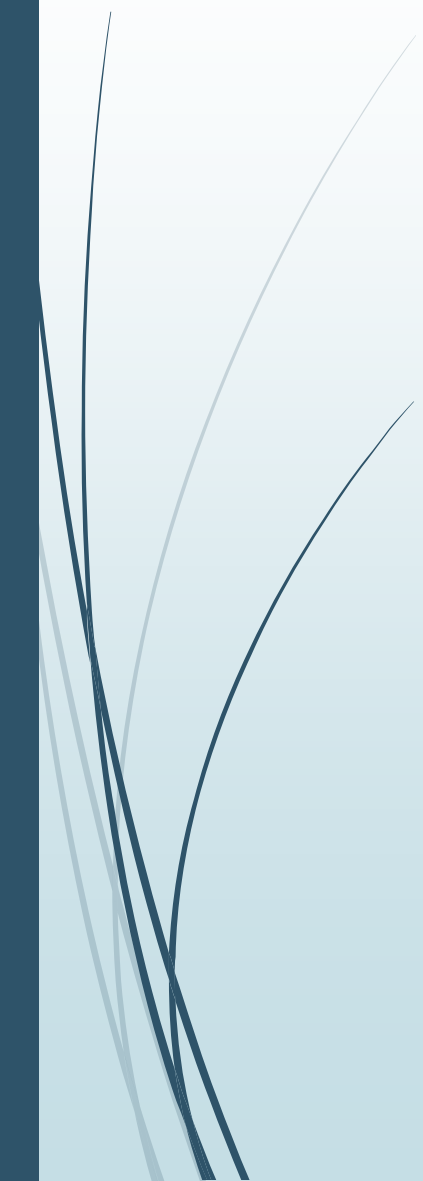


Motivation: Why Study LLM Safety?

- Unsafe outputs can lead to misinformation, bias propagation, or malicious content generation.
- Federated instruction-tuning setups amplify these risks when malicious clients inject unsafe data.
 - Inspired by Gupta et al. (2025) and Ye et al. (2024) who highlight vulnerabilities in distributed fine-tuning.



Problem Statement



- Existing safety evaluation relies heavily on large-scale models or static classifiers.
 - High computational costs make these approaches impractical for real-time or edge deployment.
 - Many safety classifiers lack interpretability and uncertainty estimation.
- Hence, there is a need for a lightweight, interpretable approach to detect unsafe content.

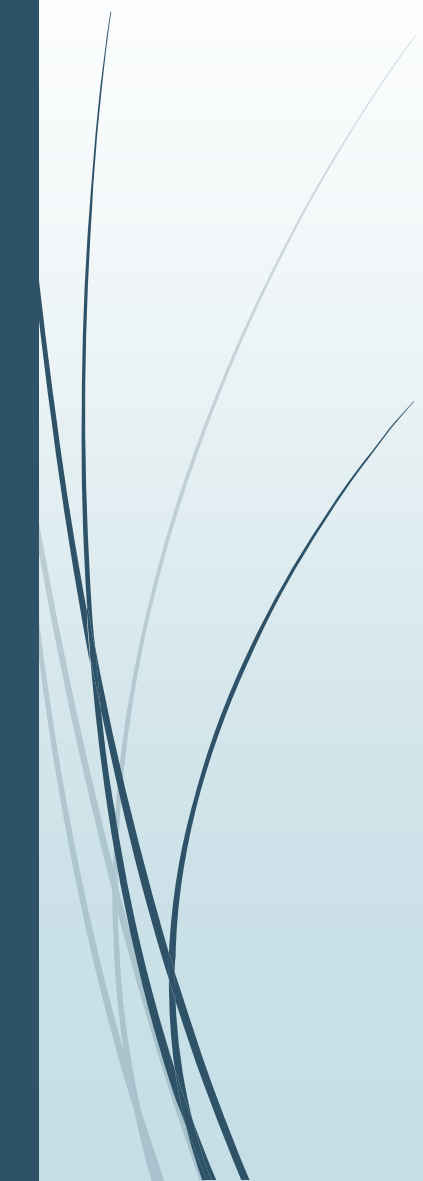


Research Gap

- Current work fails to jointly evaluate both questions and Q–A pairs for safety.
 - FedMed (Gupta et al., 2025) focuses on federated data safety but not model output safety.
 - Debate-based frameworks (Chern et al., 2024) enhance reasoning but are not tuned for safety detection.
- Gap: No study uses multiple small language models debating to classify safety at both prompt and response levels.



Research Objectives



- This research aims to:
 - Develop the Small-Language Multi-Agent Debate (SLMAD) framework for safety evaluation.
 - Enable cooperative reasoning among small models to identify unsafe content.
 - Quantify uncertainty through disagreement entropy among agents.
 - Achieve interpretability and low computation overhead.

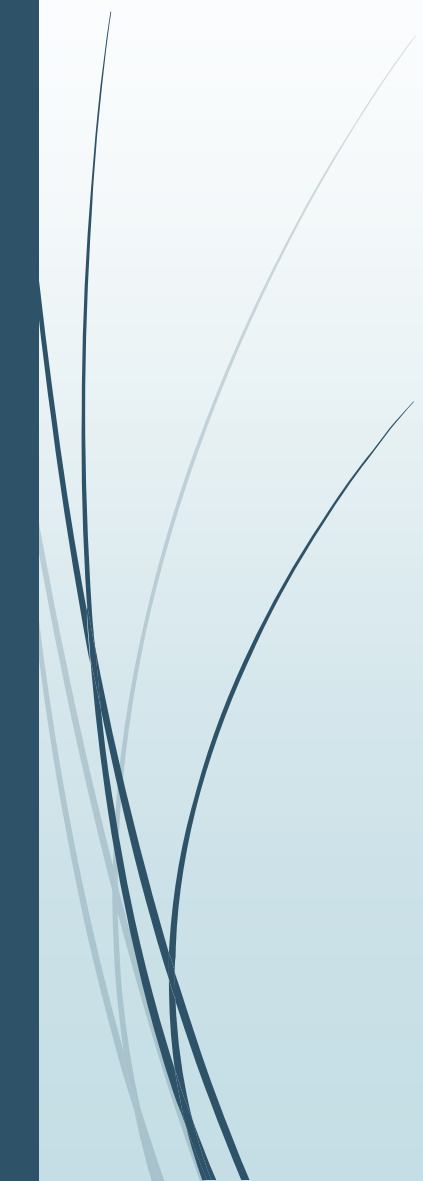


Proposed Framework: SLMAD

- Key Idea: Allow small language models to engage in structured debates to reach consensus on safety.
 - Each agent independently evaluates, critiques, and revises its judgment.
 - The moderator aggregates the results to compute a SafetyScore and uncertainty metric.



Debate Protocol



- Round 1: Independent Judgments, each agent assigns safety label, rationale, and confidence.
- Round 2: Critique & Revision, agents review others' rationales and revise their own conclusions.
- Moderator computes final SafetyScore based on weighted confidence and disagreement entropy.

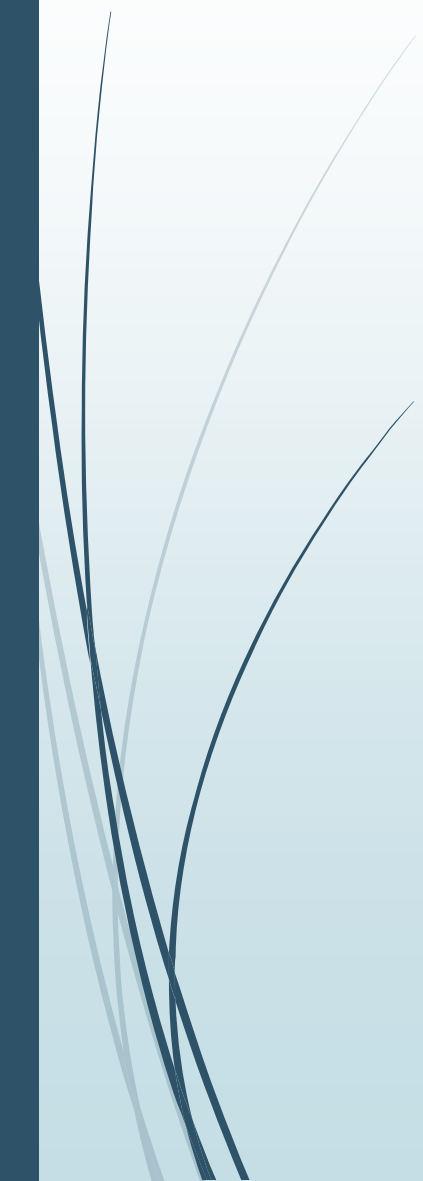


Safety Taxonomy

- SLMAD classifies data under two safety dimensions:
- Question Safety (SQ): Safe, Borderline, Unsafe.
- Q–A Safety (QA): Harmless-Refusal, Safe-Information, Unsafe-Enabling, Unclear (requires escalation).



Planned Methodology



- Data: Harmful/harmless datasets (Gupta et al., 2025), BeaverTails, AdvBench, synthetic adversarial data.
- Baselines: Single SLM classifiers, medium-scale judges, rule-based filters.
- Calibration: Platt scaling and temperature-based normalization for agent confidence.
- No experiments yet, this is a conceptual proposal stage.



Evaluation Strategy

- Performance Metrics: Accuracy, F1-score, Unsafe Recall, Disagreement Entropy, Efficiency.
- Ablation Studies: Committee size, model heterogeneity, debate depth, and confidence weighting strategies.

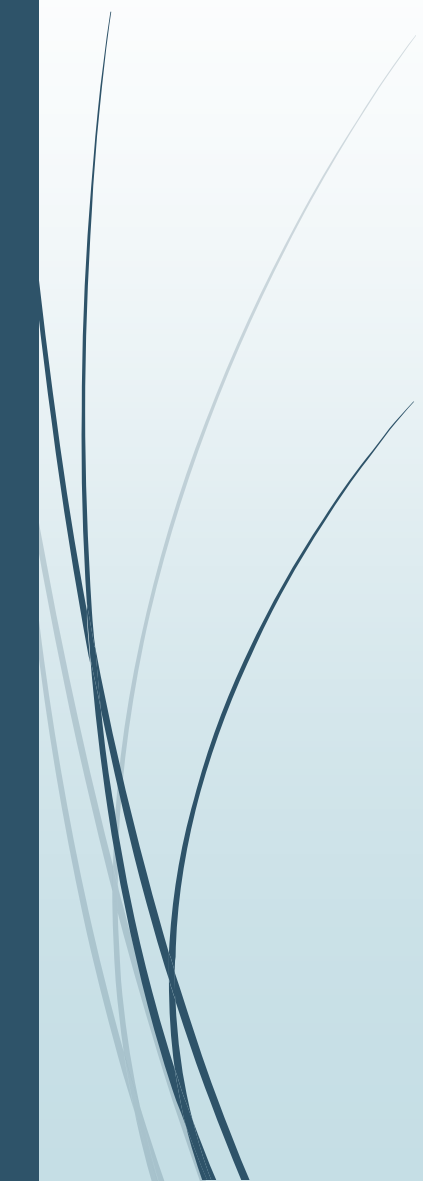


Expected Outcomes

- ▶ SLMAD is expected to:
 - ▶ Match or exceed single-model safety classifiers in reliability.
 - ▶ Provide interpretable rationales and clear uncertainty indicators.
 - ▶ Operate efficiently using small models with lower compute cost.
 - ▶ Support scalable, real-time safety assessments.



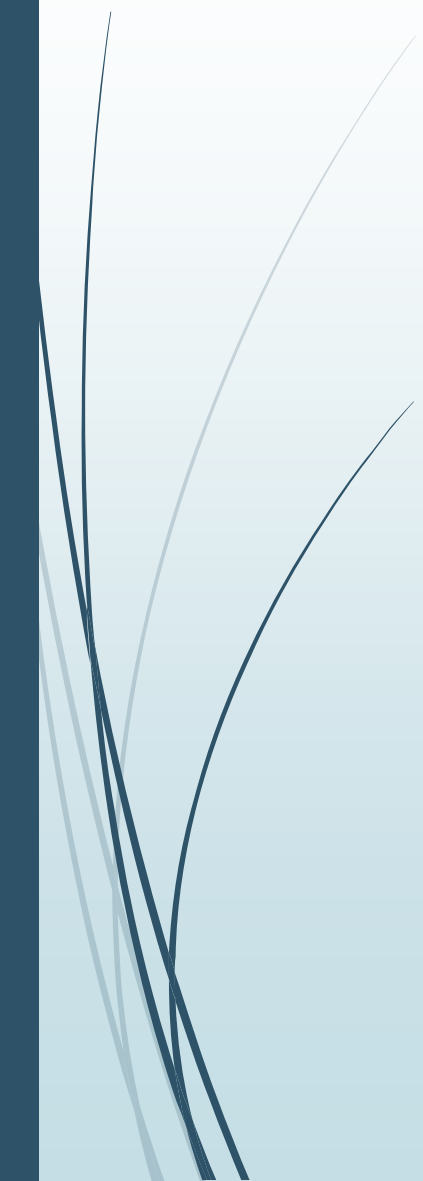
Novel Contributions



- Debate-driven safety classification among small LMs.
- Dual-level safety evaluation (question + Q–A).
- Confidence-weighted consensus mechanism for uncertainty estimation.
- A framework bridging reasoning interpretability with safety detection.

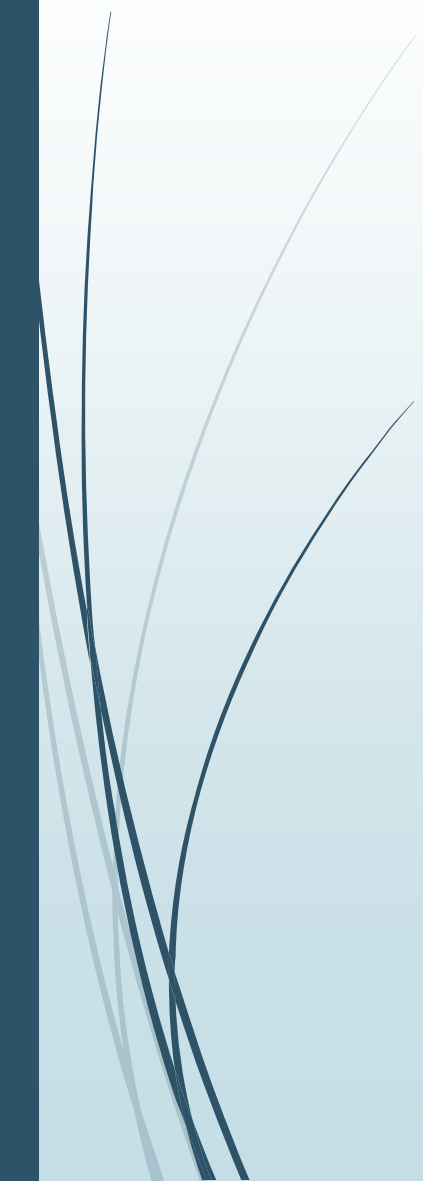


Future Work and Impact

- Integration with federated instruction-tuning frameworks.
 - Impact: Contribute to trustworthy, explainable, and scalable AI safety systems.
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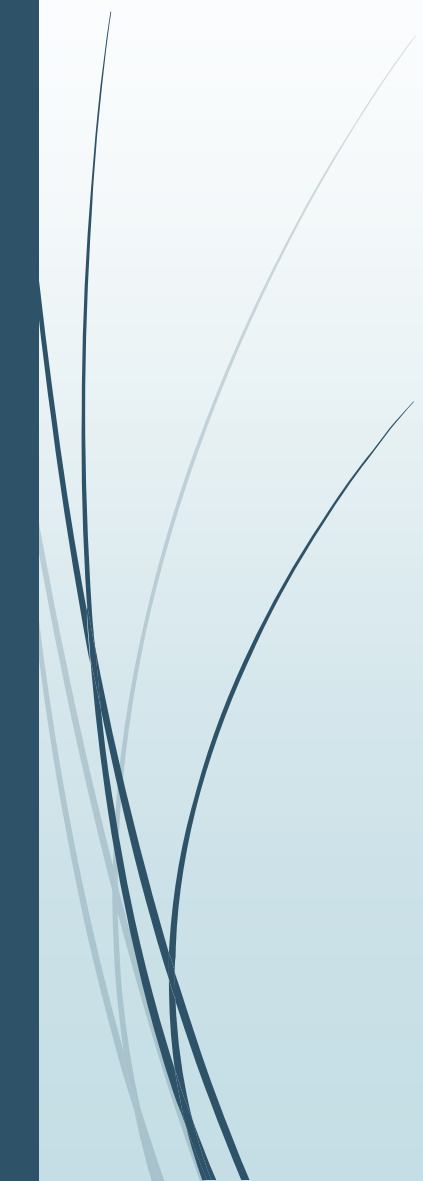
Limitations and Risks



- Requires careful design of debate prompts and rationales.
- Small models may have limited reasoning depth.
- Aggregation and calibration may introduce bias if poorly tuned.
- Extensive validation needed before real-world deployment.



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